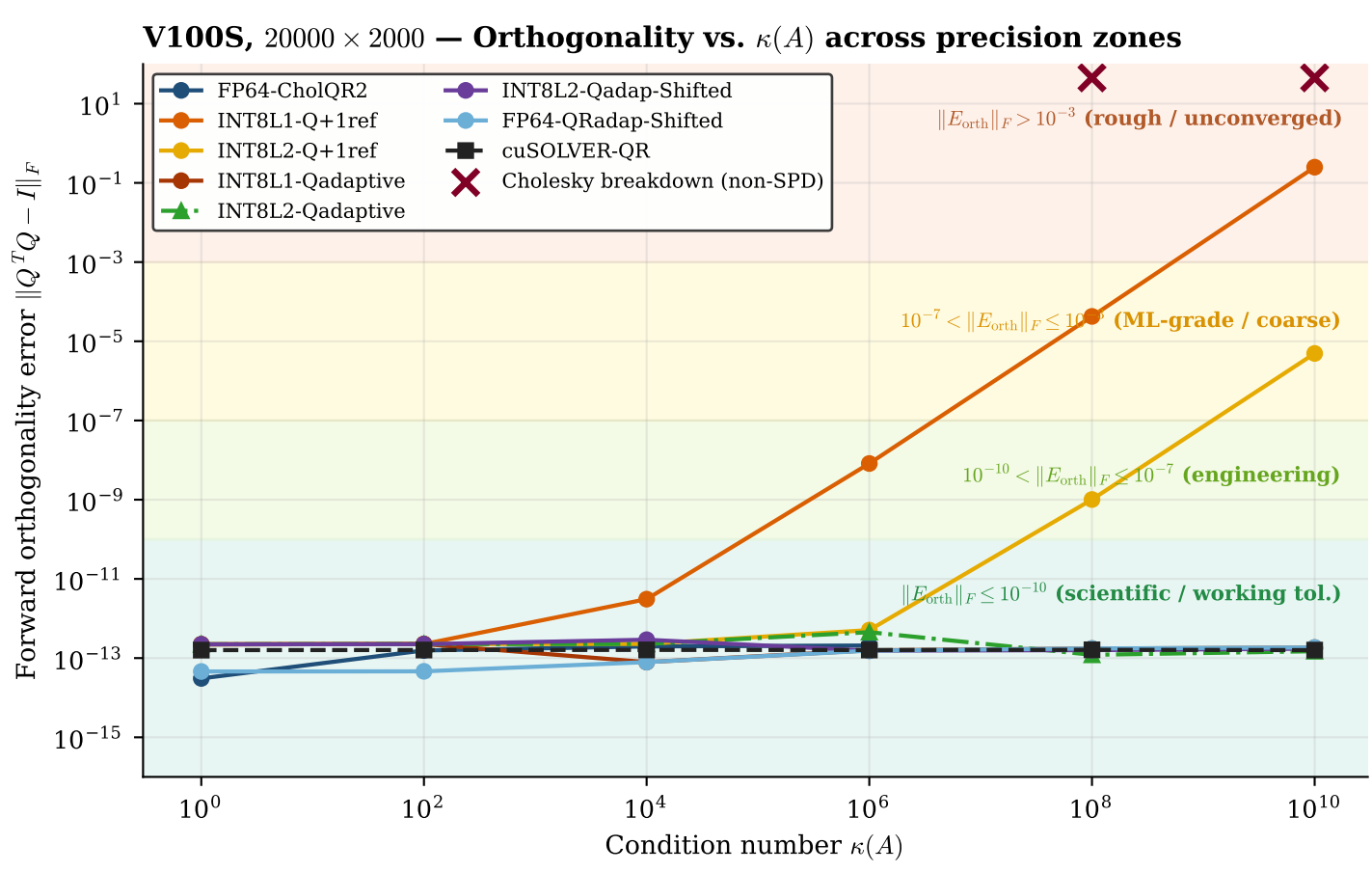
V100S, 20000 × 2000 — Orthogonality vs. κ(A) across precision zones
FP64-CholQR2
INT8L1-Q+1ref
INT8L2-Q+1ref
INT8L1-Qadaptive
INT8L2-Qadaptive
INT8L2-Qadap-Shifted
FP64-QRadap-Shifted
cuSOLVER-QR
Cholesky breakdown (non-SPD)
$\|E_{\text{orth}}\|_F > 10^{-3}$ (rough / unconverged)
$10^{-7} < \|E_{\text{orth}}\|_F \leq 10^{-3}$ (ML-grade / coarse)
$10^{-10} < \|E_{\text{orth}}\|_F \leq 10^{-7}$ (engineering)
$\|E_{\text{orth}}\|_F \leq 10^{-10}$ (scientific / working tol.)
Forward orthogonality error $\|Q^TQ - I\|_F$
Condition number $\kappa(A)$
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{0}
10^{2}
10^{4}
10^{6}
10^{8}
10^{10}